# Predicting Systemic Biological Failure in Ultramarathons Using Non-Linear Machine Learning

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## Introduction

The Western States 100-Mile Endurance Run (WSER) represents one of the most extreme tests of human physiological and biomechanical endurance. Traversing the Sierra Nevada mountains, athletes are subjected to massive cumulative elevation change and severe environmental stressors, including peak afternoon temperatures that frequently exceed 95°F in the deep canyons. Consequently, the attrition rate is significant. In this context, a “Did Not Finish” (DNF) is rarely a sudden, unpredictable event. It is typically the lagging indicator of a compounding, systemic biological failure caused by thermal, caloric, or biomechanical overload. Traditionally, medical interventions and pacing strategies in ultramarathons are entirely reactive. They treat athletes only after a catastrophic breakdown has visibly occurred.

To shift the paradigm from reactive treatment to proactive intervention, this project utilizes a longitudinal dataset comprising two decades of WSER race telemetry spanning from 2004 to 2025. This specific dataset was selected because it provides a rare, highly structured record of human performance under extreme stress. By extracting the elapsed times at unchanging geographical checkpoints, most notably Robinson Flat (mile 30) and Michigan Bluff (mile 55) as seen in Figure 1, it is possible to mathematically quantify an athlete’s pacing degradation as they navigate the most brutal sections of the course. Combining this internal biomechanical output with external environmental data, such as race-day high temperatures and regional snowpack, creates a robust feature set for prognostic modeling.

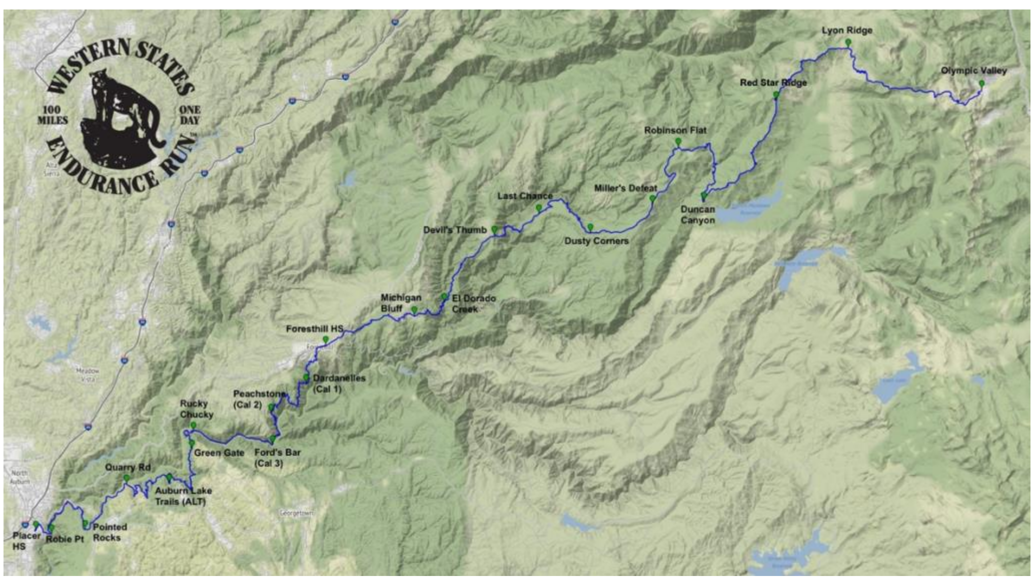


Figure 1: Course map of the 100-mile WSER going from Olympic Valley to Auburn, California [1].

The primary goal of this project is to develop and optimize a machine learning model capable of predicting systemic failure before it becomes irreversible. By comparing a linear baseline model (Logistic Regression) against a non-linear ensemble algorithm (Random Forest), this study aims to uncover the mathematical relationship between early-race pace degradation, environmental heat stress, and late-race survivability. Furthermore, the objective extends beyond pure predictive accuracy to practical clinical utility. By identifying the optimal prediction checkpoint and empirically tuning the algorithmic decision threshold, this project balances mathematical model confidence with actionable lead time. Ultimately, the goal is to transform historical race data into a real-time prognostic tool that pacers and medical teams can deploy in the field to successfully stabilize a runner.

## Methods

**Data Acquisition and Preprocessing**

The dataset was constructed using historical race results from the Western States 100 spanning the years 2004 to 2025 [2]. Because the raw data existed in heterogeneous formats including legacy HTML tables and varying Excel structures, a custom Python Extract, Transform, Load (ETL) pipeline was developed (pipe_WSER_final.ipynb). This pipeline utilized dynamic header inference and time harmonization algorithms to standardize two decades of disparate checkpoint data into a uniform matrix of continuous elapsed minutes.

To ensure the integrity of the pace degradation and split time calculations, a spatial survivorship filter was applied. Specifically, the years 2011 and 2012 were automatically excluded from the training dataset. During these years, historic snowpack forced course reroutes that bypassed the mile 30 Robinson Flat checkpoint. Because this specific aid station is required to establish a baseline pacing metric, including those years would have mathematically corrupted the training data. Finally, missing demographic values were handled using an iterative imputer, and all continuous variables were standardized using a standard scaler to ensure equal mathematical weighting during model training.

**Feature Engineering**

Machine learning models require explicit mathematical representation of biological stressors. To achieve this, raw elapsed times were engineered into specific biomechanical velocity metrics. The primary features included baseline speed (Pace Start to Robinson), physiological workload under heat stress (Pace Canyons), and the crucial system variable: Pace Degradation. This degradation feature calculated the percentage drop in speed between the relatively cool high country and the severe heat of the canyons. These internal biological outputs were then combined with external environmental inputs, specifically the forecasted high temperature in degrees Fahrenheit and the June 10$^{th}$ snowpack levels. Runner demographics (Age and Gender) were also included to complete the final feature matrix.

**Model Architecture and Optimization**

To predict systemic failure, two distinct algorithmic architectures were evaluated. A Logistic Regression model was deployed first as a linear baseline. This was used to determine if environmental and pacing stressors simply accumulate in a linear, additive fashion to cause a DNF. A Random Forest Classifier was then trained as the primary predictive engine to capture complex, non-linear biological interactions, such as how extreme heat specifically amplifies pacing degradation at different ages [3].

Because WSER has a historically high finish rate of approximately 80 percent, the training dataset was heavily imbalanced. To prevent the models from trivially predicting a finish for every runner, balanced class weights were applied during training. This mathematically penalized the algorithms more heavily for missing a DNF than for generating a false alarm, forcing the models to prioritize the minority class.

Furthermore, predictive threshold tuning was conducted to optimize the clinical utility of the Random Forest. Rather than accepting the default 50 percent decision boundary, the prediction probabilities were empirically analyzed. The threshold was shifted to find the exact mathematical percentage that maximized the model's F1-score. The F1-Score is a strict evaluation metric calculated as the harmonic mean of precision and recall. Unlike simple accuracy, the harmonic mean heavily penalizes extreme imbalances. Therefore, maximizing this score mathematically guarantees the optimal clinical trade-off between generating unnecessary false alarms (precision) and successfully catching a failing runner (recall) [4].

**Temporal and Spatial Analysis**

To determine the optimal geographical deployment for this predictive tool, a temporal loop analysis was constructed. The algorithms were sequentially trained and evaluated at three distinct physical intervention checkpoints: Robinson Flat (mile 30), Michigan Bluff (mile 55), and Foresthill (mile 62). This analysis mapped the intersection of algorithm confidence against the remaining race distance to identify the exact aid station that provides the highest prediction accuracy while leaving enough mileage for a pacer or medical team to successfully execute an intervention.

During the development of this project, Google's Gemini 3.1 Pro large language model was utilized to troubleshoot Python syntax errors, optimize the machine learning pipeline, and refine the grammatical structure of the final report [5].

## Results

**Baseline vs. Complex Performance**

The predictive performance of the algorithms was evaluated using a test dataset of runners who reached the mile 62 checkpoint. The Logistic Regression model achieved a high Recall score of 0.72 for the DNF class. This indicates it successfully identified 72% of failing runners. However, it suffered from a high false alarm rate, resulting in a low DNF Precision score of 0.25 and an overall system accuracy of 67%.

The Random Forest model sacrificed a minor amount of Recall to massively improve predictive precision. Figure 2 details the confusion matrices for both algorithms, demonstrating how the non-linear model successfully reduced false alarms while increasing the overall system accuracy to 81%.

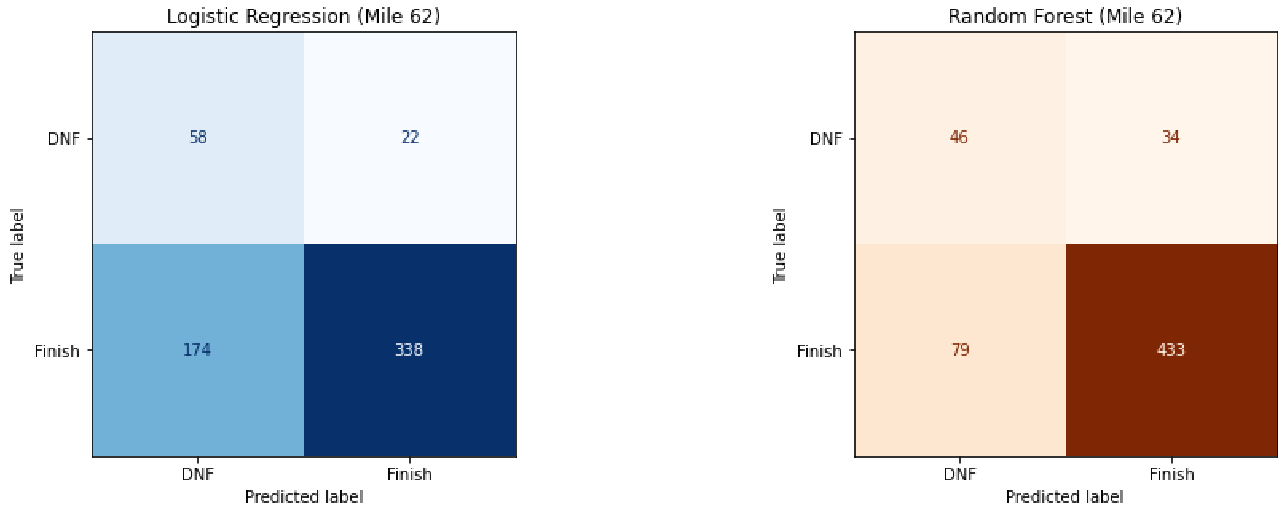


Figure 2: Confusion matrices comparing the predictive accuracy of the Logistic Regression baseline and the Random Forest algorithm at mile 62.

**The Driver of Systemic Failure**

To understand the physiological mechanics of a DNF, the feature weights of both models were extracted and compared. Figure 3 visualizes the directional drivers for the linear model alongside the absolute predictive power of the non-linear model. The Logistic Regression confirms the hypothesis that severe environmental thermal stress (forecasted high temperature) linearly subtracts from a runner's probability of survival. However, the Random Forest reveals that environmental inputs are secondary to biomechanical outputs. Pacing degradation through the canyons dictates the Random Forest's predictions. The heat itself does not exclusively cause the DNF. Instead, the physiological consequence of the heat, measured as pacing degradation triggers systemic failure.

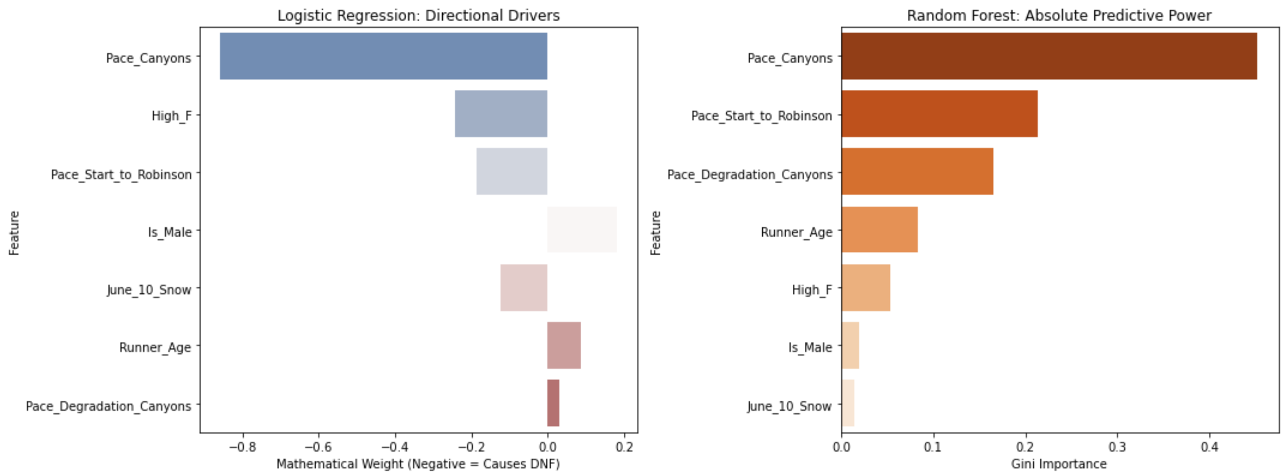


Figure 3: Feature importances illustrating the mathematical weight of environmental and biomechanical variables in predicting a DNF.

**Threshold Optimization**

By default, the classification algorithms trigger a DNF prediction at a 50% confidence interval. Figure 4 illustrates the performance metrics across all possible confidence thresholds for the Random Forest model. The optimal intervention point occurs at a threshold of 0.63. By requiring the algorithm to be 63% confident before flagging a runner, the system maximizes the F1-Score to optimize the trade-off between Precision and Recall.

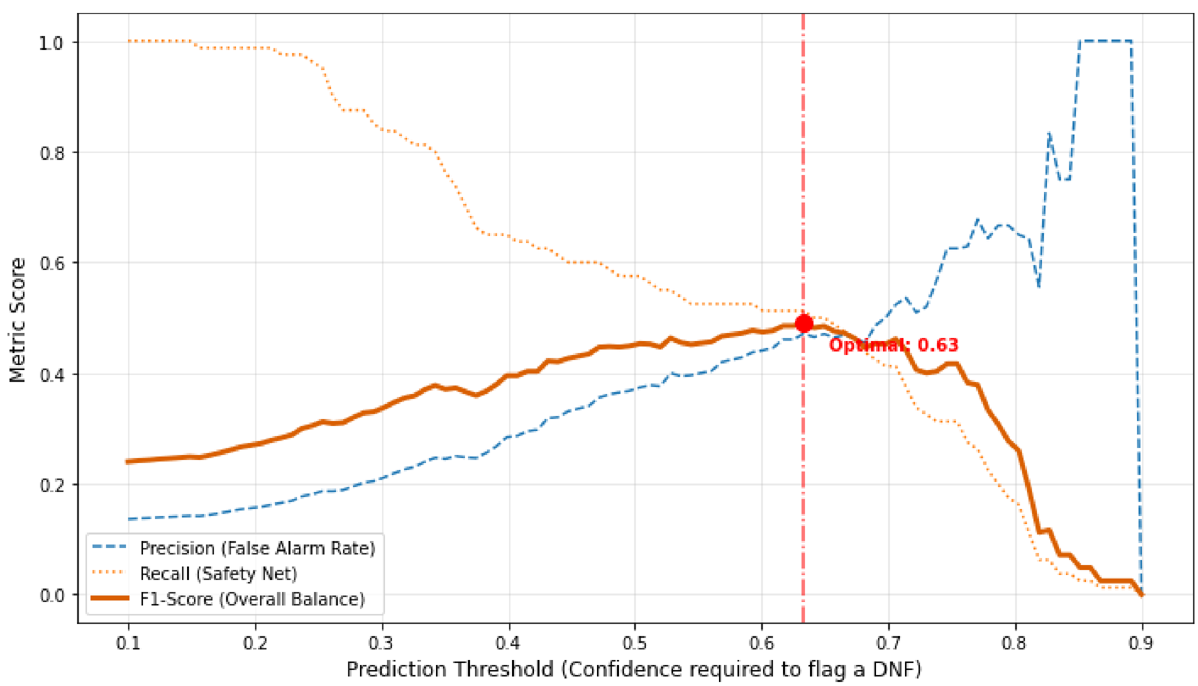


Figure 4: Threshold optimization curve demonstrating the peak F1-Score intersection at a 63% prediction confidence.

**Temporal Prediction Analysis**

To determine the optimal geographical deployment for the prediction engine, the models were tested chronologically at mile 30, 55, and 62. Figure 5 maps the predictive accuracy of the algorithms against the actionable lead time remaining in the race. Predictions at mile 30 offer over 69 miles of intervention time but suffer from exceptionally low accuracy because runners have not yet faced the thermal stress of the canyons. Conversely, waiting until mile 62 yields only marginal gains in accuracy while sacrificing critical intervention time. Michigan Bluff at mile 55 represents the optimal intersection. It captures the massive biomechanical data from the canyons while preserving over 44 miles of actionable runway for a medical team to intervene.

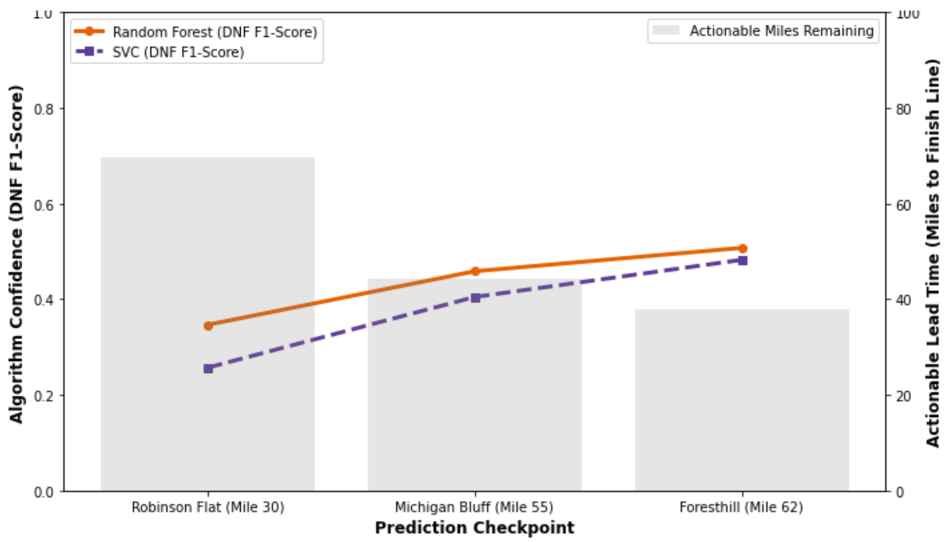


Figure 5: The actionability trade-off curve identifying Michigan Bluff (mile 55) as the optimal prediction checkpoint.

**Model Strengths and Weaknesses**

The primary strength of the final predictive model is its reliance on longitudinal, immutable checkpoint data combined with an empirically derived intervention threshold. This makes it a highly objective tool for medical teams. However, the system has inherent weaknesses. It relies solely on external tracking telemetry and cannot account for critical unrecorded variables, such as a runner's precise caloric intake, hydration status, or psychological willpower. Because human biology and determination are inherently volatile, the model will always exhibit a moderate margin of error. Additionally, ultramarathon environments include a nearly unexplainable level of complexity in relation to athlete performance, with influences from past races and race seasons, competitive racing dynamics, and logistical mishaps, such as not receiving support from crew at a location when it is expected.

## Conclusion

This project successfully developed a proactive machine learning tool to predict systemic biological failure in the Western States 100. The study demonstrated that a non-linear Random Forest ensemble significantly outperforms standard linear baseline models. The feature extraction phase revealed a critical physiological insight: internal biomechanical output, specifically pacing degradation through the canyons, is a far superior predictor of a DNF result than raw ambient heat. Furthermore, prognostic trade-off analysis mathematically optimized the clinical utility of the model. By shifting the predictive intervention threshold to 63% and deploying the algorithm at the Michigan Bluff checkpoint (mile 55), the system achieves maximum predictive confidence while preserving over 44 miles of actionable lead time. Ultimately, this initial analysis barely skims the surface of the complexity that endurance athletes face in 100-mile footraces but begins to provide a framework for medical teams and pacers to transition from reactive treatment to proactive intervention.

## References


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